POWER_INPUT — Rev A authoritative capture
TPS259470A protection + fixed 3.3V TPS62132



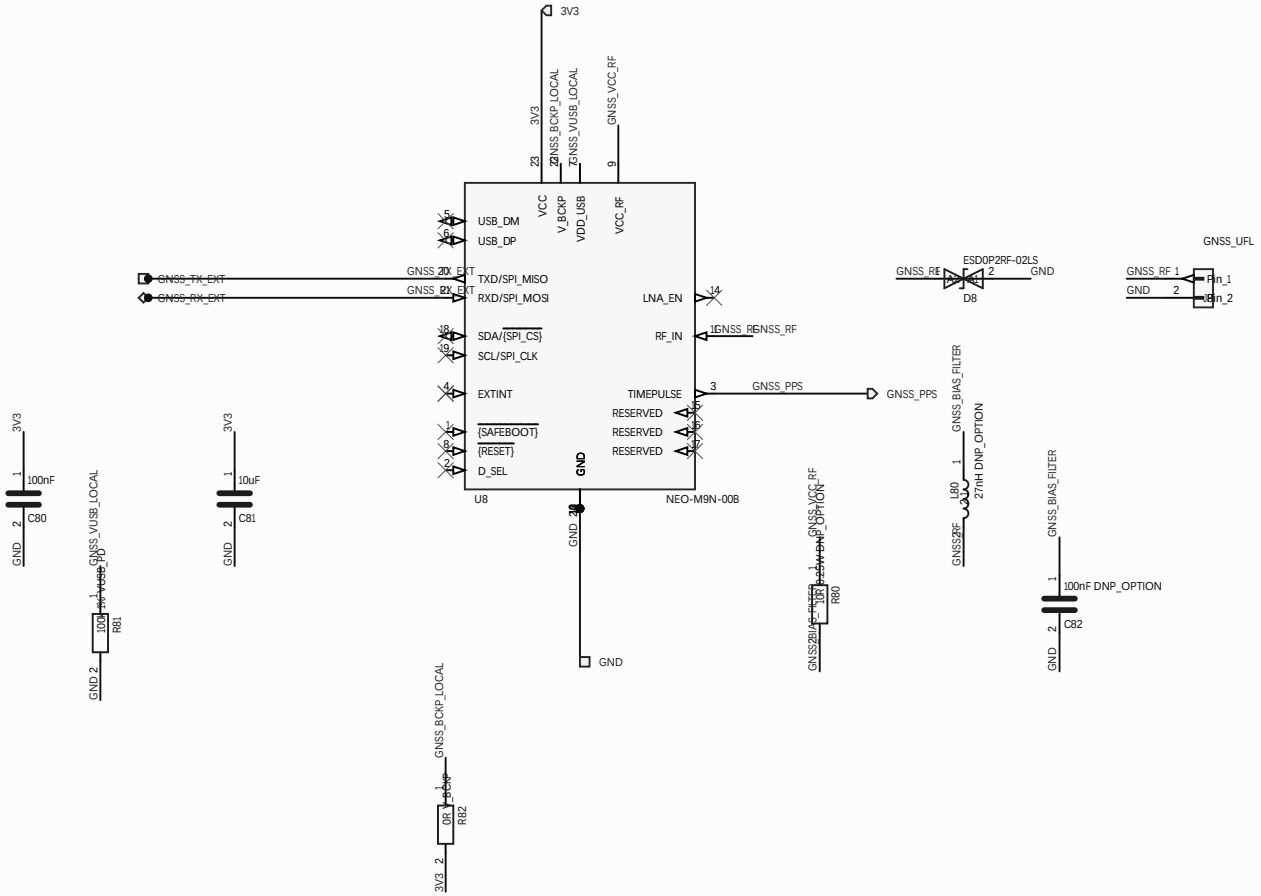
Sheet: /		
File: 10_POWER_INPUT.kicad_sch		
Title: EVT_SYS_CARRIER		
Size: A4	Date:	Rev:
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

MCU_CORE — STM32H753IIT6
P0 time capture pins remain fixed by PINMAP_SYS_001.csv

Y1
ECS-2520S33-250-FN-TR
BLM88D60SH1
TC2030_SWD
DEBUG_TX
TP1
DEBUG_RX
TP2
U1
STM32H753IIT6
GND

Sheet: /		
File: 20_MCU_CORE.kicad_sch		
Title: EVT_SYS_CARRIER		
Size: A3	Date:	Rev:
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

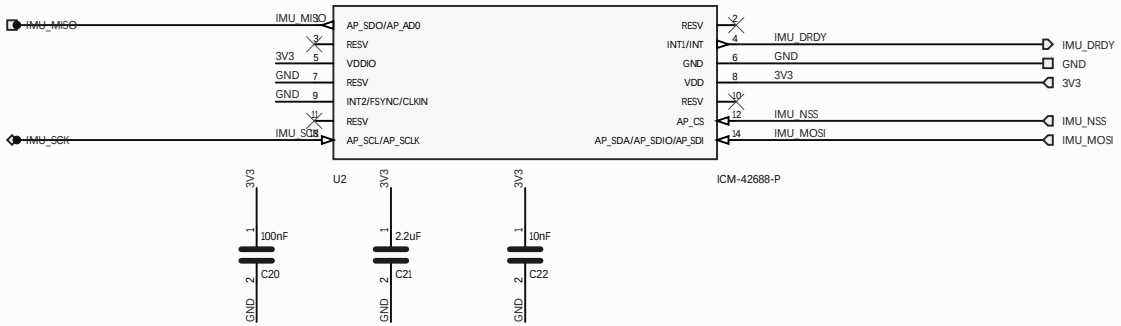
TIME_GNSS — NEO-M9N timing and RF entry
P0 GNSS_PPS -> STM32 PA0/TIM2_CH1



R82 ties V_BCKP to 3V3 for EVT. R81 pulls unused V_USB to GND.
L80/R80/C82 implement the optional active-antenna VCC_RF bias-T.

Sheet: /		
File: 30_TIME_GNSS.kicad_sch		
Title: EVT_SYS_CARRIER		
Size: A4	Date:	Rev:
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

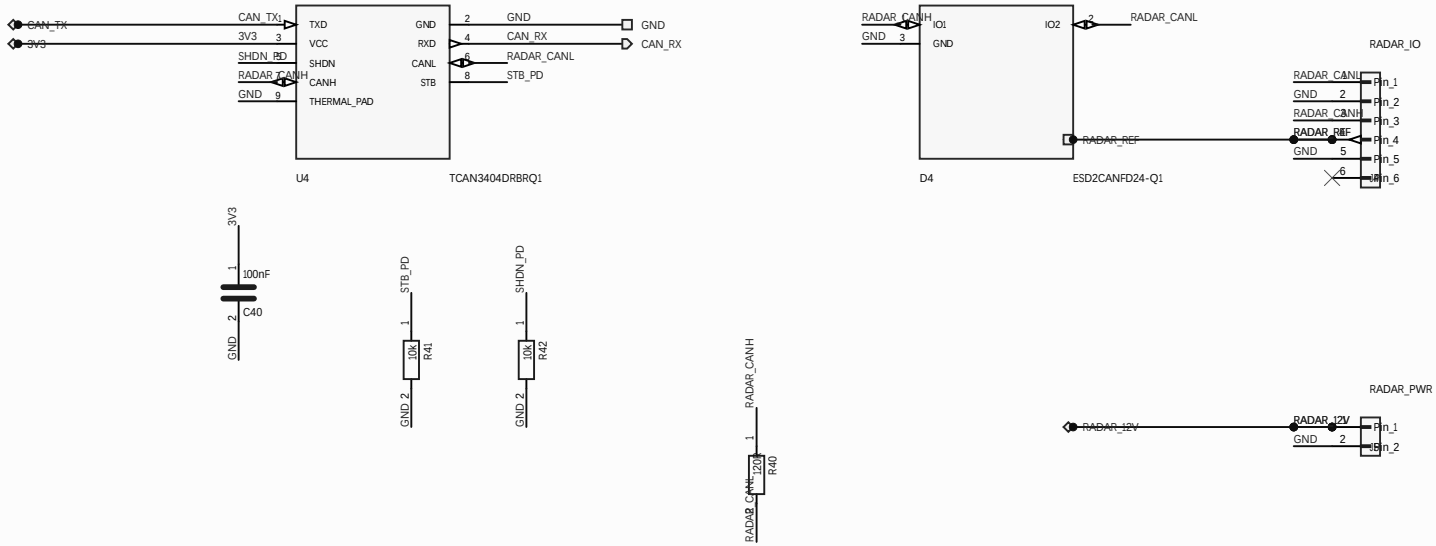
IMU — ICM-42688-P SPI2 + hardware DRDY
P0 IMU_DRDY -> STM32 PB11/TIM2_CH4



C20/C21 are VDD decoupling; C22 is the dedicated VDDIO 10nF bypass.
Pin7 RESV and pin9 FSYNC are tied to GND per the Rev.A pin contract.

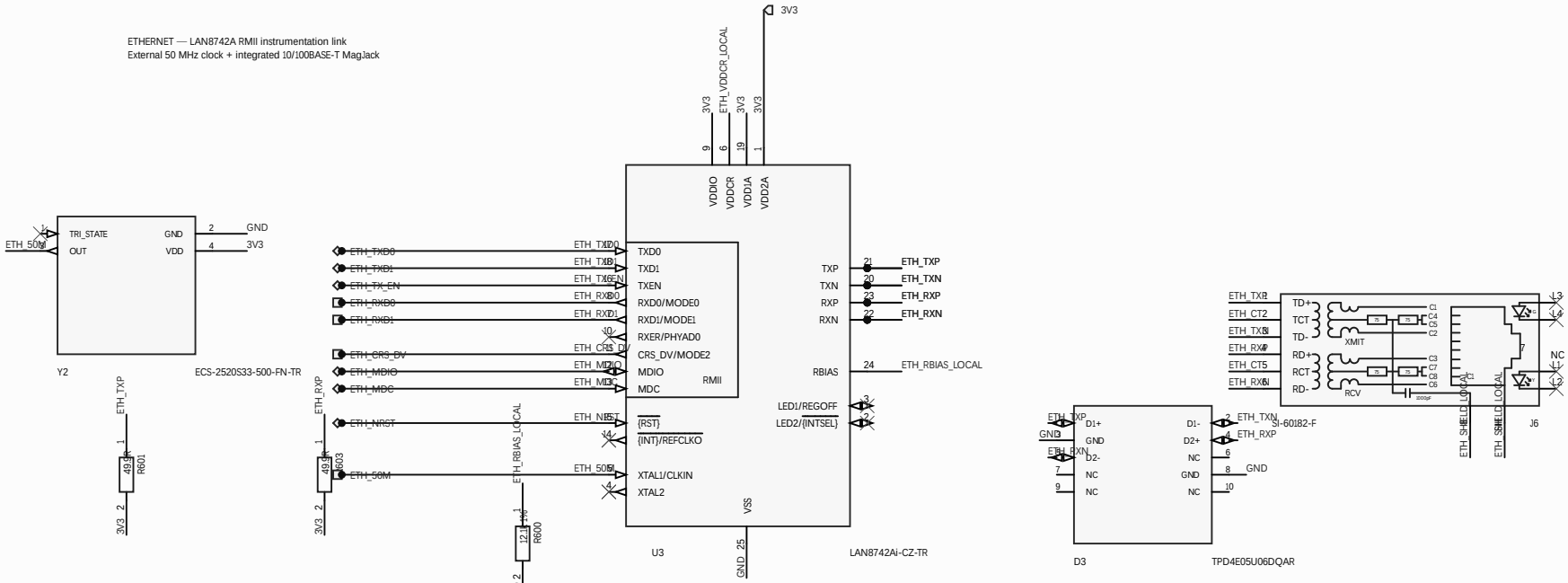
Sheet: /		
File: 40_IMU.kicad_sch		
Title: EVT_SYS_CARRIER		
Size: A4	Date:	Rev:
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

RADAR_IO_CAN — external AWR2944-class EVM interface
CAN-FD control/metadata + P0 RADAR_REF timestamp + protected 12V



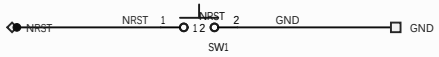
R40 120R 1% is DNP by default; populate only at the CAN endpoint.
R41/R42 10k 1% force normal mode. J4 pin6 remains NC by Rev.A contract.

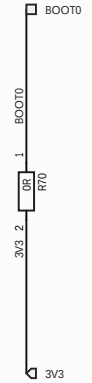
Sheet: /		
File: 50_RADAR_IO_CAN.kicad_sch		
Title: EVT_SYS_CARRIER		
Size: A4	Date:	Rev:
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1



Sheet: /	
File: 60_ETHERNET.kicad_sch	
Title: EVT_SYS_CARRIER	
Size: A4	Date:
KiCad E.D.A. kicad (5.1.4)-1	Rev:
	Id: 1/1

DEBUG_BOOT — SWD, reset and boot service access
No vehicle actuation/control interfaces





Primary Tag-Connect SWD connector remains captured with MCU_CORE.
SW1 pulls NRST low; R70 is a DNP service override only.

Sheet: /		
File: 70_DEBUG_BOOT.kicad_sch		
Title: EVT_SYS_CARRIER		
Size: A4	Date:	Rev:
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

EVT System Carrier
External radar front-end: TI AWR2944PEVM
Carrier contains no 76-81 GHz RF routing

POWER_INPUT
3V3
RADAR_12V
SYS_EFUSE_FLT
RADAR_EFUSE_FLT
GND
10_POWER_INPUT.SchDoc
GND
3V3
RADAR_12V
SYS_EFUSE_FLT
RADAR_EFUSE_FLT

ETH_NRST
ETH_MDIO
MCU_CORE
BOOT0
RADAR_EFUSE_FLT
SYS_EFUSE_FLT
ETH_RXD1
ETH_RXD0
ETH_CRS_DV
ETH_50M
CAN_RX
GNSS_TX_EXT
IMU_MISO
IMU_DRDY
RADAR_REF
GNSS_PPS
3V3
20_MCU_CORE.SchDoc
GND
IMU_NSS
IMU_SCK
IMU_MOSI
GNSS_RX_EXT
CAN_TX
ETH_MDC
ETH_TX_EN
ETH_TXD0
ETH_TXD1
ETH_NRST

TIME_GNSS
GNSS_PPS
GNSS_TX_EXT
GND
3V3
GNSS_RX_EXT
30_TIME_GNSS.SchDoc
GND
GNSS_PPS
GNSS_TX_EXT

IMU
IMU_DRDY
IMU_MISO
IMU_MOSI
IMU_SCK
IMU_NSS
3V3
40_IMU.SchDoc
GND
IMU_DRDY
IMU_MISO

RADAR_IO_CAN
CAN_RX
RADAR_REF
RADAR_12V
CAN_TX
3V3
50_RADAR_IO_CAN.SchDoc
GND
CAN_RX
RADAR_REF

ETH_MDIO
ETHERNET
ETH_MDIO
ETH_NRST
ETH_TXD1
ETH_TXD0
ETH_TX_EN
ETH_MDC
3V3
60_ETHERNET.SchDoc
GND
ETH_50M
ETH_CRS_DV
ETH_RXD0
ETH_RXD1

NRST
DEBUG_BOOT
NRST
GND
BOOT0
3V3
70_DEBUG_BOOT.SchDoc
GND
BOOT0
3V3

Sheet: /
File: EVT_SYS_CARRIER.kicad_sch
Title: EVT_SYS_CARRIER
Size: A2
Date:
Rev:
KiCad E.D.A. kicad (5.1.4)-1
Id: 1/1